

NODE Platform

Hardware Reference · 8-Channel Addressable LED Controller Family

Wateefy Electronics

The power section is the product. The module is a consumable.

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Scope

This document covers the hardware common to every board in the NODE family — power architecture, input protection, output stage, environmental data and build process. It applies to **NODE Mini**, **NODE Flex** and **NODE Backbone** alike.

What differs between variants — the MCU module, the logic rail source, board dimensions, GPIO mapping and per-board accepted limitations — is documented in each board's own **Variant Specification**. Where a variant departs from anything stated here, its supplement says so explicitly in *Deltas from the Platform Reference*.

Read this document for how the board protects your pixels. Read the supplement for which board to buy.

1. Power Architecture

Three separate domains, each on its own copper region:

Domain	Boundary	What it carries
Input (unprotected)	Screw terminal → ideal-diode stage	Raw supply. Assumed hostile. Transient clamp sits across this node.
Clean (protected)	Ideal-diode output → main fuse	Reverse-blocked, surge-clamped. Bulk capacitance lives here.
Distribution bus (fused)	Main fuse → 8 channel fuses + logic regulator	Everything downstream taps this bus through its own fuse.

2. Input & Protection

Parameter	Value	Notes
Input voltage range	5 – 48 VDC	Single rail, powers both pixels and onboard logic
Input connector	WECO 158-A-111/02, 2-pole screw terminal	57 A / 10 mm² rated; 30 A design maximum = 53% of rating
Maximum continuous current	30 A	At standard 1 oz outer copper — no copper upgrade required
Main fuse	30 A mini blade, Keystone 3568 holder	Holder rated 30 A / 500 V; user-replaceable
Reverse-polarity protection	LM74700-Q1 ideal-diode controller + CSD18540Q5B	FET held off under reverse input; no current path to load; non-sacrificial and self-recovering
Pass element	CSD18540Q5B NexFET	60 V V_{DS} , 205 A I_D , ~2.2 mΩ $R_{DS(on)}$ — 15% of current rating at 30 A design load
Input transient clamp	SMCJ48CA bidirectional TVS, 1500 W	48 V standoff, 53.3 V breakdown; conducts in neither direction at rated input
Bulk capacitance	2 × 1000 µF, 63 V	On the clean rail, downstream of the ideal diode
Ideal-diode charge pump	22 nF, 100 V	VCAP to ANODE, floating — not referenced to ground
Ideal-diode hold-up	47 µF electrolytic, 80 V	Between controller and pass FET, per reference design
Polarity indication	Anti-parallel green/red LED pair	Green = correct, red = reversed; shared 1 kΩ limit; LED reverse stress 2 V against 5 V rating

Voltage headroom summary: 36 V nominal, 48 V absolute maximum. Every component in the power path is rated **63 V or above**; every semiconductor is **60 V or above**.

Derating against input voltage. Utilization of each component's own rating. **5 V, 12 V and 24 V are the mainstream pixel voltages** and are shown in bold:

Component	Rating	5 V	12 V	24 V	36 V	48 V
Bulk input capacitors	63 V	8 %	19 %	38 %	57 %	76 %
Channel output caps	80 V	6 %	15 %	30 %	45 %	60 %

Component	Rating	5 V	12 V	24 V	36 V	48 V
Ideal-diode hold-up	80 V	6 %	15 %	30 %	45 %	60 %
Buck input electrolytic	80 V	6 %	15 %	30 %	45 %	60 %
HV-plane ceramics	100 V	5 %	12 %	24 %	36 %	48 %
Ideal-diode controller	65 V	8 %	18 %	37 %	55 %	74 %
Pass MOSFET V_{DS}	60 V	8 %	20 %	40 %	60 %	80 %
Buck regulator V_{IN}	80 V	6 %	15 %	30 %	45 %	60 %

At the **36 V nominal** rating no part on the plane exceeds 60 % of its own limit. At the **48 V absolute maximum** nothing exceeds 80 %. On the 5 V, 12 V and 24 V strips that carry the overwhelming majority of installations, the entire plane runs under 40 %.

3. Outputs

Parameter	Value	Notes
Channels	8	Independent, individually fused — no shared fuses
Output connector	Phoenix MKDS-series 3-pole, per channel	V+ / DATA / GND; 12 A rated → 63% at the 7.5 A fitted fuse, 83% at the 10 A maximum
Per-channel fuse	7.5 A mini blade fitted, 10 A maximum — Keystone 3568 holder	25% of holder rating at 7.5 A, 33% at 10 A; user-replaceable
Level translation	SN74HCT541 octal buffer, DIP-20 socket	3.3 V drive vs 2.0 V V_{IH} = 1.3 V margin
Data series resistance	330 Ω , axial through-hole, per channel	Limits fault current back into the buffer
Data ESD protection	PESD5V0F1BL, per channel, connector side	0.4–0.55 pF junction capacitance; RC $\approx$ 181 ps — no measurable effect on WS281x bit timing
Data drive current	$\approx$ 4 mA per channel	67% of $\pm$ 6 mA buffer rating
Buffer supply current	$\approx$ 45 mA total	64% of 70 mA device maximum
Per-channel decoupling	47 μ F electrolytic 80 V + 100 nF ceramic 100 V	Local to each output connector
Channel activity indication	One LED per channel, 470 Ω limited	$\approx$ 5 mA, 25% of LED rating

4. Environmental & Compliance

Parameter	Value
Operating temperature	−20 °C to +60 °C (<i>design target; not qualification tested</i>)
Substrate rating	UL-recognized, JLC-1 marking
RoHS	Compliant
Workmanship	IPC Class 2
Electrical test	Flying-probe, 100% of boards
Traceability	2D barcode serial number, per board

Mounting & Enclosure

No enclosure is included, and none is planned. This is the norm in this segment — bare board, mounted in whatever the installation already uses: a CG1500, a DIN rail, a weatherproof box, a plywood backer.

What NODE adds instead is **conformal coating**, applied after assembly with connectors, sockets, fuse clips, and switches masked. Competing boards in this class ship uncoated. A coated board in a vented enclosure tolerates condensation and humidity in a way a bare board does not, which matters for the seasonal outdoor installs this was built for.

Four corner mounting holes are provided. *Hole spacing and a downloadable mount STL: to be published at first article.*

Build & Assurance

- **Fabrication** — 4-layer, 1.6 mm, high- T_g substrate, ENIG finish, epoxy-filled and capped vias, IPC Class 2 workmanship
- **Assembly** — professional SMT assembly, flying-probe electrical test on 100% of boards

- **Traceability** — 2D barcode serial on every board; UL-recognized substrate marking
 - **Coating** — conformal coating applied after assembly, with connectors, sockets, fuse clips, and switches masked
 - **Design review** — every net verified against manufacturer datasheets; schematics independently reviewed by two automated analysis passes with human verification of every finding
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